

An internal inconsistency in TLMC conjecture 00000001672

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Abstract

The conjecture identifies 77 as the Zarankiewicz formula's value at $(7, 7)$ and asserts both $\text{cr}(K_{7,7}) = 77$ and that the formula holds for $\min(m, n) \leq 7$. The formula's value is actually 81. Consequently the headline and its asserted restatement cannot both hold, regardless of the true crossing number. We formalize this contradiction for every natural-valued function in Lean 4, without defining crossing numbers or using any axioms. Independently, Woodall's published computer-assisted theorem gives $\text{cr}(K_{7,7}) = 81$ and $\text{cr}(K_{7,9}) = 144$, refuting the isolated headline as well. That crossing-number theorem is cited, not formalized here. The description of $K_{7,7}$ as the smallest open instance is also false; the smallest unsettled cases are $K_{7,11}$ and $K_{9,9}$.

1 Statement and interpretation

The English statement of conjecture **00000001672** is reproduced verbatim below, with its mathematical notation typeset:

Definition: The Zarankiewicz formula asserts $\text{cr}(K_{m,n}) = \lfloor m/2 \rfloor \lfloor (m-1)/2 \rfloor \lfloor n/2 \rfloor \lfloor (n-1)/2 \rfloor$. Conjecture: $\text{cr}(K_{7,7}) = 77$ (the value of the formula at the smallest open instance); that is, the Zarankiewicz formula holds for $\min(m, n) \leq 7$.

The source is the repository file `conjectures/00000001672.md` [1].

For positive integers m, n , let $K_{m,n}$ be the complete bipartite graph with part sizes m, n . Its crossing number is the minimum number of edge crossings in a plane drawing. Write the displayed product as

$$Z(m, n) = \left\lfloor \frac{m}{2} \right\rfloor \left\lfloor \frac{m-1}{2} \right\rfloor \left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor.$$

The initial “Definition” specifies the formula under discussion; it does not supply a proof that the formula holds for crossing numbers.

For a function $c : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$, distinguish the three claims in the text:

$$\begin{aligned} H(c) : & \quad c(7, 7) = 77, \\ V : & \quad Z(7, 7) = 77, \\ F(c) : & \quad \forall m, n \in \mathbb{N}, \quad \min(m, n) \leq 7 \implies c(m, n) = Z(m, n). \end{aligned}$$

We read the declarative headline and the phrase “that is” as asserting both $H(c)$ and $F(c)$; the parenthesis additionally asserts V . This preserves the numerical headline, the exact product, the boundary 7, and the universal range of the restatement. We prove that $H(c) \wedge F(c)$ is impossible, even after omitting V , and separately that V is false. Thus the entire collection of assertions is inconsistent.

Graph parameters are usually positive. The formalization also proves the same impossibility when F quantifies only over $m, n \geq 1$. Only $(7, 7)$ is used, so the convention for zero-sized parts and the natural subtraction at $m = 0$ or $n = 0$ cannot affect the contradiction. For positive parameters, natural division by 2 is precisely the floor in the displayed expression.

2 Full mathematical proof

Lemma 1. $Z(7, 7) = 81$, and in particular $Z(7, 7) \neq 77$.

Proof. Both $\lfloor 7/2 \rfloor$ and $\lfloor (7-1)/2 \rfloor = \lfloor 6/2 \rfloor$ equal 3. Therefore

$$Z(7, 7) = 3 \cdot 3 \cdot 3 \cdot 3 = 81 \neq 77.$$

□

Theorem 2. For every $c : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$,

$$\neg(c(7, 7) = 77 \wedge \forall m, n, \min(m, n) \leq 7 \implies c(m, n) = Z(m, n)).$$

The same conclusion holds if the universal quantifiers are restricted to positive integers.

Proof. Suppose both assertions hold. Since $\min(7, 7) = 7 \leq 7$, the second assertion gives $c(7, 7) = Z(7, 7) = 81$. The first gives $c(7, 7) = 77$. Transitivity yields $77 = 81$, a contradiction. The witness $(7, 7)$ has both parameters positive, which proves the positive-parameter variant with exactly the same argument. □

In particular,

$$F(c) \implies c(7, 7) \neq 77.$$

Even the weaker local assertion $c(7, 7) = Z(7, 7)$ contradicts the headline. All these conclusions are independent of the definition or actual value of the crossing number. Instantiating $c(m, n)$ with the actual crossing number is a legitimate mathematical specialization of the theorem; that specialization requires no crossing-number estimates.

3 Scope and other readings

Headline alone. Woodall’s computer-assisted theorem establishes $\text{cr}(K_{7,7}) = 81$, so the isolated headline $\text{cr}(K_{7,7}) = 77$ is false [3]. This is a cited mathematical refutation; Woodall’s crossing-number computation is not formalized in this submission. The Lean proof independently refutes the headline and the asserted formula range taken jointly, without depending on that computation.

Formula range alone. The assertion that the formula holds for all $\min(m, n) \leq 7$ is a separate claim. The Lean arithmetic argument neither proves nor refutes that range for the actual crossing number. Woodall settles the instances $(7, 7)$ and $(7, 9)$, not the full range: $K_{7,11}$ remains among the smallest unsettled cases. Replacing 77 by 81 repairs the headline and agrees with Woodall’s theorem, but does not establish the universal range.

Bare logical equivalence. If “that is” is interpreted as the single formula $H(c) \leftrightarrow F(c)$ without asserting either side, the universal negation of this equivalence is false. Both sides can be false. For example, let $c(m, n) = 0$ for all m, n . Then $H(c)$ is false because $0 \neq 77$, and $F(c)$ is false because at $(7, 7)$ it would require $0 = 81$. Hence $H(c) \leftrightarrow F(c)$ is true for this abstract function. Lean formally exhibits this example; the zero function is not proposed as the crossing-number function. For the actual crossing number, Woodall makes H false and the *local* formula assertion $\text{cr}(K_{7,7}) = Z(7, 7)$ true, so the local equivalence is false. This does not settle F , which asserts the formula for an entire range, or the global bare equivalence $H \leftrightarrow F$.

Literature context. Kleitman’s 1970 result establishes the Zarankiewicz formula for complete bipartite graphs with smaller part at most 6 [2]. Woodall, in *Cyclic-order graphs and Zarankiewicz’s crossing-number conjecture*, *Journal of Graph Theory* **17** (1993), 657–671, proved by a computer-assisted search that [3]

$$\text{cr}(K_{7,7}) = 81, \quad \text{cr}(K_{7,9}) = 144$$

Thus $K_{7,7}$ is a solved case, and both the isolated headline and its description as the “smallest open instance” are false. The smallest unsettled cases are $K_{7,11}$ and $K_{9,9}$. These crossing-number results are cited, not formalized here, and are not premises of the Lean proof. The formal result is the inconsistency of the stated assertions, including $Z(7, 7) = 81 \neq 77$, for every function c .

4 Lean statement and semantic correspondence

The file `lean4/ZarankiewiczConsistency.lean` begins with the exact statement and the interpretation formalized. It uses no imports, only core Lean natural arithmetic. The symbol `Nat` is Lean’s type of natural numbers; the source supplies a local notation for that type. Automatic implicit parameters are disabled.

The definition and main theorem, written using `Nat` and ASCII notations equivalent to those in the source, are:

```
def Z (m n : Nat) : Nat :=
  (m / 2) * ((m - 1) / 2) * (n / 2) * ((n - 1) / 2)

theorem conjecture_inconsistent (cr : Nat -> Nat -> Nat) :
  Not (cr 7 7 = 77 /\
    forall m n, min m n <= 7 -> cr m n = Z m n)
```

The fully qualified name is `ZarankiewiczConsistency.conjecture_inconsistent`. The curried function `cr` represents c above. It has no imposed properties, so the result covers any natural-valued crossing-number function. The conjunction faithfully retains the headline and the full range assertion, rather than introducing a surrogate graph invariant. The proof specializes the universal quantifiers at 7, 7.

The additional theorems are:

- `Z_seven`: $Z(7, 7) = 81$.
- `not_value_of_formula`: $Z(7, 7) \neq 77$.
- `local_inconsistent`: the headline and the local formula equality cannot both hold.
- `formula_range_excludes_77`: $F(c) \Rightarrow c(7, 7) \neq 77$.
- `positive_parameters_inconsistent`: the main contradiction with the additional premises $1 \leq m$ and $1 \leq n$ in the range assertion.
- `equivalence_reading_has_abstract_model`: there exists a natural-valued function satisfying the bare equivalence; the witness is the zero function described in Section 3.

No assertion about the actual crossing number is hidden in these auxiliary theorems. Every claim in the proof of Theorem 2, including the arithmetic evaluation, is covered by the formalization.

5 Verification, axioms, and reproduction

The pinned toolchain is `leanprover/lean4:v4.33.0`. The project declares `Mathlib v4.33.0`; its manifest pins revision `db584cd6d46c92f209a44c0f1c829460d327499d`. Although `Mathlib` is declared in the project configuration, the proof needs only Lean’s core library. The project has one library root and one proof file.

Every theorem listed above ends with an axiom audit command. The actual compiler output in `verification.txt` reports that all seven theorems do not depend on any axioms, including:

```
'ZarankiewiczConsistency.conjecture_inconsistent'  
does not depend on any axioms
```

The source contains no `sorry`, `admit`, `native_decide`, or new axiom declarations. Closed arithmetic goals use the ordinary kernel-checked `decide` tactic. There are no unproved assumptions, graph-computation certificates, or external numerical dependencies in the main theorem.

The final proof file compiled successfully with Lean v4.33.0. The compiler output and exit status are saved in `verification.txt`. A fresh project can be built with:

```
cd lean4  
lake exe cache get && lake build
```

The independent sanity check `verify.py` uses only Python's standard library. It computes rational floors exactly, checks all four factors, checks $Z(7, 7) = 81 \neq 77$, and checks that $(7, 7)$ lies in the specified range. It neither computes nor assumes a crossing number. Its actual output is appended to `verification.txt`. To reproduce the supplementary check and report from the submission directory:

```
python3 verify.py  
pdflatex -interaction=nonstopmode -halt-on-error report.tex  
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Attribution. Submitted by AlyciaBHZ on behalf of the Omega Institute (trureturing project, <https://github.com/the-omega-institute/trureturing>). No code from trureturing or another submission is vendored.

References

- [1] The Last Math Competition, conjecture 00000001672, English statement.
<https://github.com/The-Last-Math-Competition/The-Last-Math-Competition/blob/main/conjectures/00000001672.md>.
- [2] D. J. Kleitman, The crossing number of $K_{5,n}$, *Journal of Combinatorial Theory* **9**(4) (1970), 315–323. [doi:10.1016/S0021-9800\(70\)80087-4](https://doi.org/10.1016/S0021-9800(70)80087-4).
- [3] D. R. Woodall, Cyclic-order graphs and Zarankiewicz's crossing-number conjecture, *Journal of Graph Theory* **17**(6) (1993), 657–671. [doi:10.1002/jgt.3190170602](https://doi.org/10.1002/jgt.3190170602).